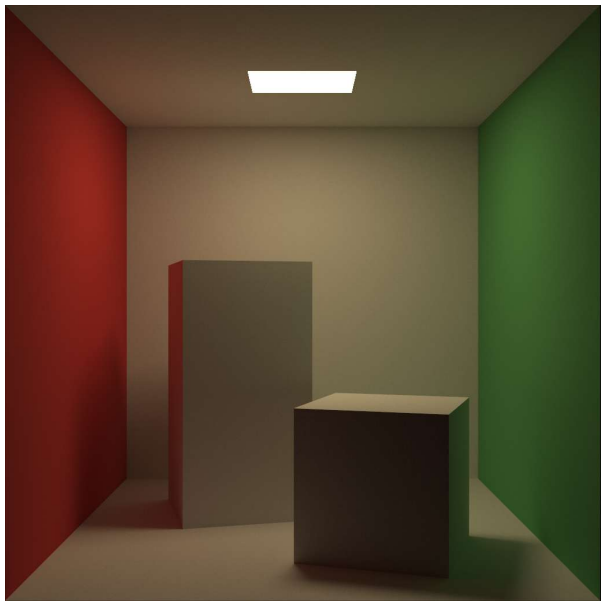
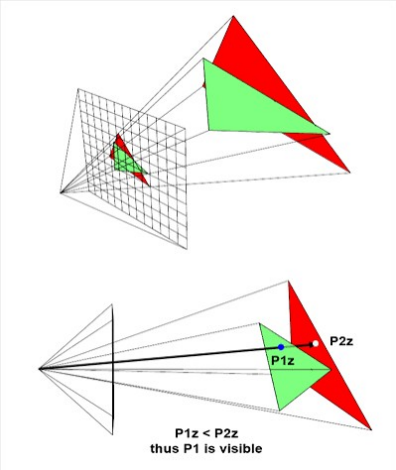
Raytracing Workshop



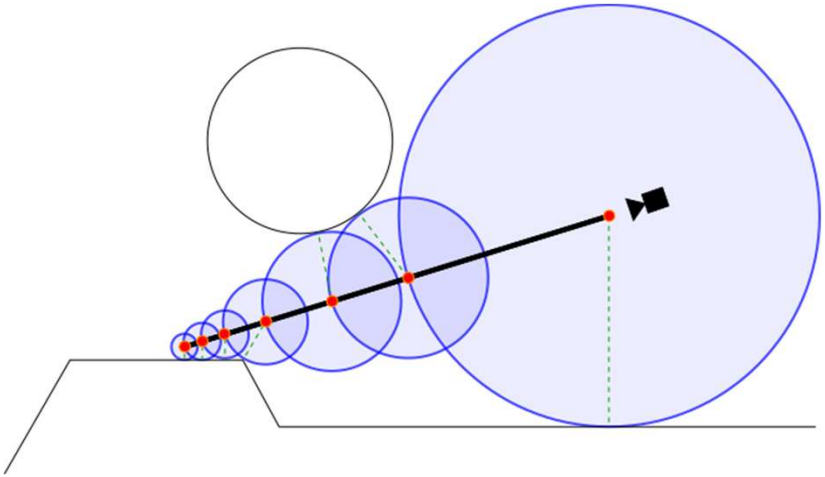
Motivation

- How do we render a 3d scene to a 2d image?

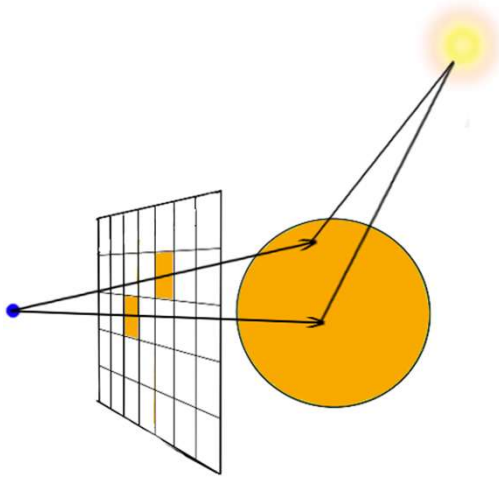
Rasterization



Ray Marching



Raytracing



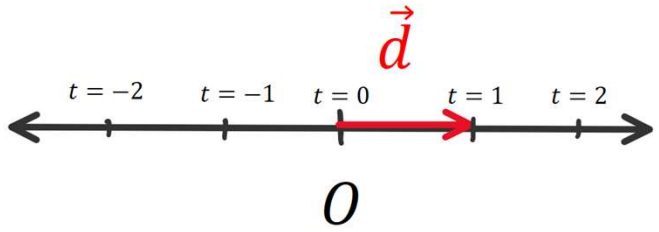
How do we simulate light?

- Treat individual photons as a ray
- Rays come from light sources with an intensity and a color
- Rays bounce around the scene and “collect” color
- Rays hit the camera sensor
- Realistic, but very slow



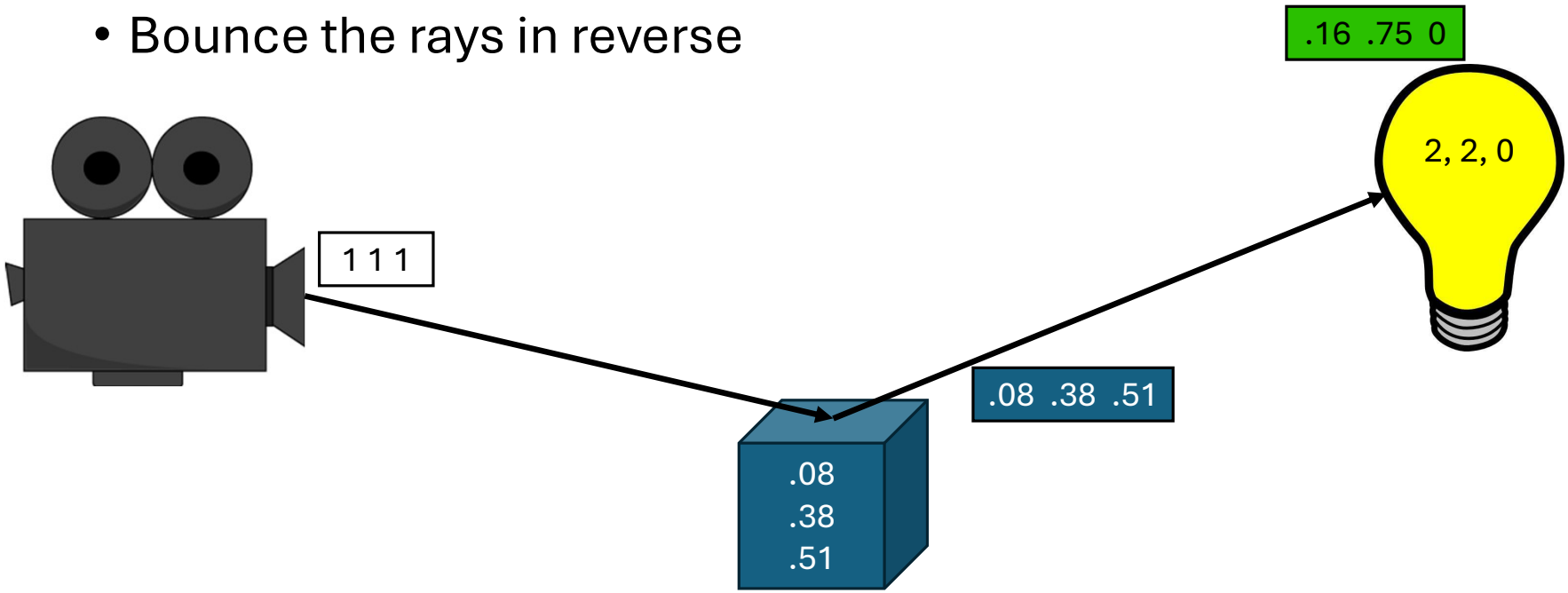
How do we represent rays?

- With an origin $\vec{O}$ and direction $\vec{d}$
- Rays start at the origin and extend infinitely in their direction
- A time value t gives a position along the ray
- $R(t) = O + t\vec{d}$



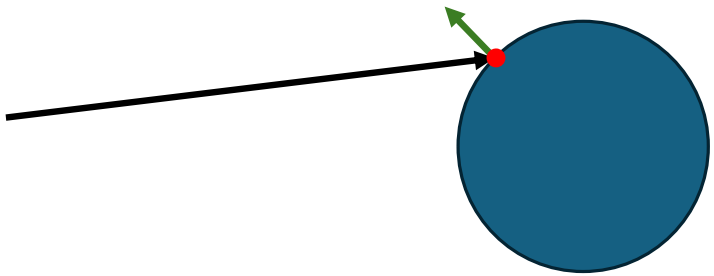
Make it faster

- Multiplying RGB components approximates “collecting” light
- Multiplication is communitive
- Bounce the rays in reverse



How do rays bounce?

- Unlike ray marching, rays do not “move forward”
- We find the intersection points analytically
- Different formula for each shape in the scene
- Produces the time and point of intersection
- Produces the normal vector at that point



Ray – Sphere intersection

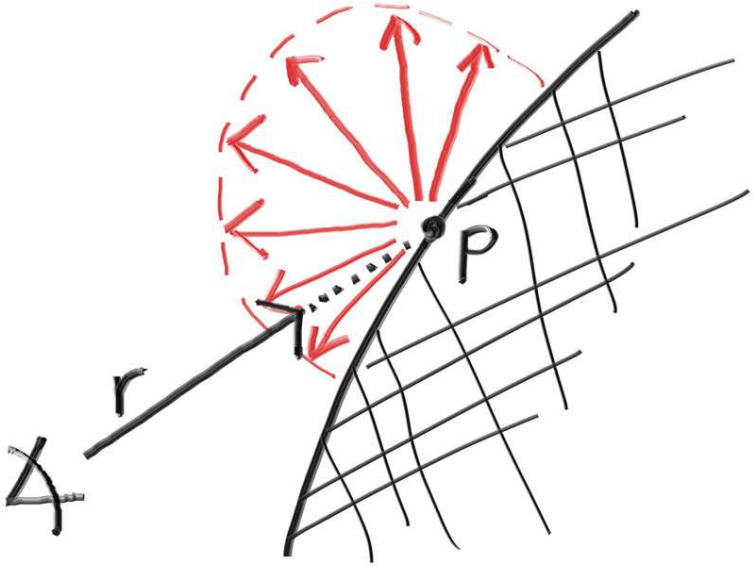
- todo

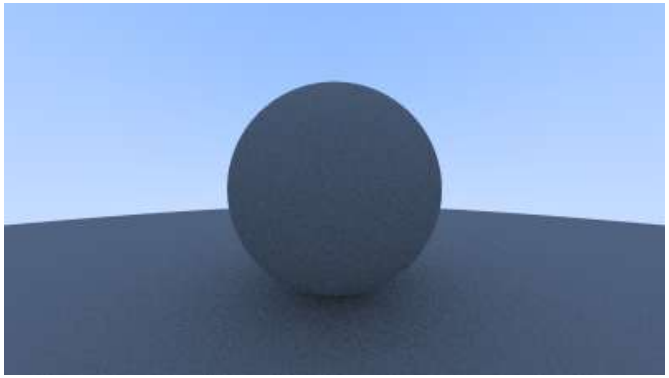
Ray scattering

- After a ray hits a surface, we need to scatter it
- Influences the material properties of the surface
- Diffuse scattering
- Lambertian scattering
- Reflective scattering
- Dielectric scattering (next week)

Diffuse scattering

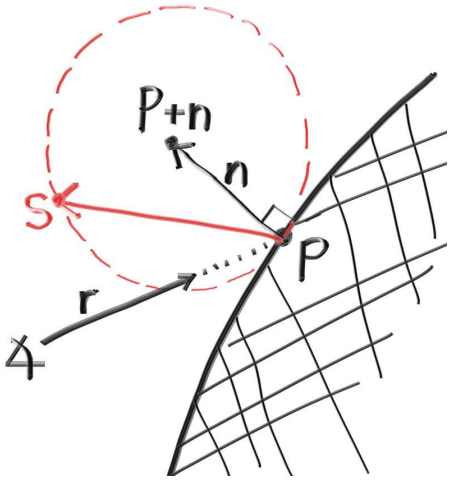
- Mimics rough, non-reflective surfaces
- A ray is bounced in a random direction in the same hemisphere as the normal

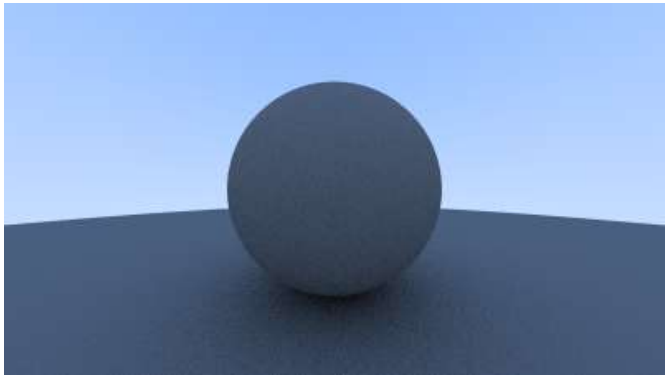




Lambertian scattering

- Similar to diffuse scattering
- Rays are more likely to scatter near the normal





Reflective scattering

- Used to mimic metals, mirrors, and other reflective surfaces

